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NF-κB in cancer: a marked target

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Abstract

The imbalance between proliferation and programmed cell death (apoptosis) is one of the critical cellular events that lead to oncogenesis. While there is no doubt that uncontrolled cell proliferation is essential for the development of cancer, deregulation of apoptosis may play an equally important role in this process. Inhibition of apoptosis prevents the death of tumor cells with DNA damage either associated with carcinogenic initiation or cancer therapy. The transcription factor NF-κB is a key regulator in oncogenesis. By promoting proliferation and inhibiting apoptosis, NF-κB tips the balance between proliferation and apoptosis toward malignant growth in tumor cells.

Keywords: Cancer; NF-κB; Transcription; Anti-apoptosis; Therapy

1. Introduction

For a long time, oncogenesis has been viewed as the result of unlimited proliferation of tumor cells, which is certainly at the core of this process [1]. Many oncogenes were identified by virtue of their ability to induce uncontrolled cell proliferation [1]. Proto-oncogenes or tumor suppressor genes are often the components of signaling pathways involved in proliferation or cell division, thereby stimulating cell growth [1–3]. Compelling evidence shows that the other side of the coin, programmed cell death (apoptosis), is likely to play an equally important role in oncogenesis [4–6]. The transcription factor NF-κB is a key regulator of immune responses and inflammation operating through the induction of numerous genes, including those coding for cytokines, chemokines and adhesion molecules [7,8]. NF-κB may also be involved in oncogenesis. Many oncogenes can activate NF-κB, whose activity is required for subsequent transformation [9,10]. Furthermore, the viral counterpart of NF-κB, v-Rel, is highly oncogenic [11]. This notion is consistent with the discovery that NF-κB induces expression of cell cycle regulators such as cyclin D1 [9]. The analysis of NF-κB deficient mice and cells led to the identification of a novel function for this versatile transcription factor—the inhibition of apoptosis [12]. As shall be discussed later, the anti-apoptotic function of NF-κB is tightly linked to its oncogenic activity.

Several recent reviews have described the role of NF-κB in oncogenesis with respect to its function in promoting cell proliferation and transformation [9,10]. In this review, we will begin by briefly reviewing the signaling pathways that lead to NF-κB activation and then focus on the role of NF-κB as an anti-apoptotic regulator in oncogenesis.

2. The pathways to NF-κB activation

NF-κB belongs to the Rel family, which contains five mammalian Rel/NF-κB proteins: RelA (p65), c-Rel, RelB, NF-κB1 (p50/p105) and NF-κB2 (p52/100) [8,13]. While RelA, c-Rel and RelB are synthesized as matured proteins, p50 and p52 are first synthesized as large precursors p105 and p100, respectively, which are processed by the proteosome. The activity of NF-κB is controlled by shuttling from the cytoplasm to the nucleus in response to cell stimulation. NF-κB dimers containing RelA or c-Rel are retained in the cytoplasm through interaction with the inhibitors of NF-κB (IκBs). In response to a variety of stimuli, IκBs are phosphorylated (Ser32 and Ser36 for IκBα and Ser19 and Ser21 for IκBβ) by the activated IκB kinase (IKK) complex, followed by rapid ubiquitin-dependent degradation by the 26S proteosome [7]. This allows NF-κB dimers to translocate to the nucleus, where they stimulate expression of target genes. IKK is composed of two catalytic subunits, IKKα and IKKβ (also known as IKK1 and IKK2), and an essential regulatory

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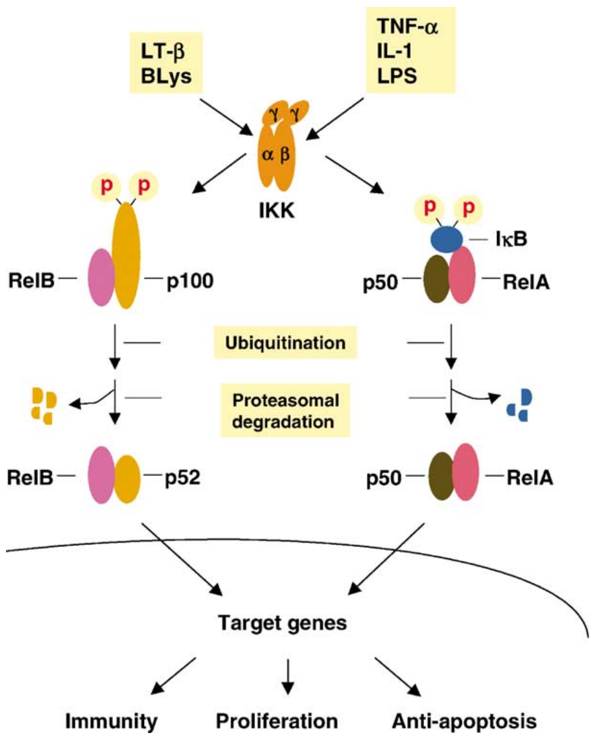

Fig. 1. Two distinct IKK signaling pathways that lead to activation of NF-κB transcription factors. The canonical pathway is activated by TNFα, IL-1 or byproducts of bacterial and viral infections (such as LPS and double stranded RNA). This pathway utilizes IKKβ to phosphorylate IκB, triggering its ubiquitin-dependent degradation and resulting in activation of RelA:p50. The second pathway, stimulated by only a few members of the TNF family, such as lymphotoxin β (LTβ) and BLys (also known as BAFF), involves IKKα, which is responsible for phosphorylation dependent processing of p100 and activation of RelB:p52. P, phosphate.

subunit, IKKγ (also known as NEMO) (Fig. 1). While IKKβ is mostly required for the canonical NF-κB pathway that depends on IκB degradation [14–16], IKKα is involved in a non-canonical NF-κB pathway that regulates, at least, the RelB/p52 dimer [17]. In resting cells, RelB is associated with p100 in the cytoplasm. Upon cell stimulation, the IκB-like C terminus of p100 is proteolyzed, resulting in RelB-p52 dimers that translocate to the nucleus [18]. Activation of the canonical NF-κB pathway is essential for innate immunity [17], whereas activation of the non-canonical pathway is involved in lymphoid organ development and adaptive immunity [17]. Once in the nucleus, NF-κB dimers are further modified, mostly through phosphorylation of the Rel proteins, to fully upregulate their transcriptional activity. This process involves protein kinase A, phosphatidyl inositol 3 kinase (PI3K)/Akt and TNF-α signaling pathways [19–21].

3. An anti-apoptotic transcription factor

The first indication that NF-κB can act as an anti-apoptotic transcription factor came from analysis of *RelA*$^{-/-}$ knockout mice, which die at embryonic day (E) 15 due to massive apoptosis of hepatocytes in the liver [22]. *RelA*$^{-/-}$ fibroblasts also exhibited enhanced sensitivity to pro-apoptotic stimuli such as tumor necrosis factor α (TNF-α). This is particularly interesting since, despite its reputation, TNF-α is a poor inducer of apoptosis unless the synthesis of RNA or protein is inhibited [23,24]. The importance of NF-κB's anti-apoptotic function was independently demonstrated by dissection of the different signaling pathways activated by TNF-α via its type 1 receptor (TNFR1) [25]. Although TNF-α activates caspases that are responsible for initiation and execution of apoptosis, the concurrent activation of NF-κB ablates TNF-α's pro-apoptotic activity. This is because certain NF-κB target genes code for inhibitors of caspase activation and apoptosis (see the description later). The suppression of apoptosis by NF-κB is an important component of TNF-α biology. The mid-gestational lethality and massive liver apoptosis of *RelA*$^{-/-}$ mice, which also occur in mice deficient in IKKβ or IKKγ [14,15,26], are completely suppressed by the absence of TNFR1 or TNF-α [27,28]. The long-term survival of lymphoid cells also requires NF-κB activity [17,29]. In addition to suppression of TNF-α-induced apoptosis, NF-κB inhibits apoptosis induced by a variety of DNA-damaging chemotherapeutic drugs and ionizing radiation in epithelial cells [9,30]. As many tumors, both of lymphoid and epithelial origin, exhibit constitutive NF-κB activity, inhibition of NF-κB may be critical for successful cancer therapy (see the description later).

4. Mechanisms beyond caspase inhibition

NF-κB suppresses apoptosis by inducing expression of a number of genes whose products inhibit apoptosis, including inhibitors of apoptosis (IAPs), caspase 8-FADD-like IL-1β-converting enzyme (caspase 8-FLICE) inhibitory protein (cFLIP), A1 (also known as Bfl1), TNF receptor associated factor 1 (TRAF1) and TRAF2. These anti-apoptotic proteins work in a coordinated fashion to block apoptosis at multiple steps along the apoptotic cascade or regulate other pro- or anti-apoptotic pathways (Fig. 2).

The cellular IAPs (cIAPs) are the most extensively studied NF-κB-induced anti-apoptotic proteins. By directly binding and inhibiting caspase 3 and caspase 7, as well as preventing activation of pro-caspase 6 and pro-caspase 9 [31], cIAPs inhibit both extrinsic and intrinsic apoptotic pathways. The best example is that NF-κB may inhibit TNF-α-induced apoptosis through induction of cIAP1 and cIAP2, along with TRAF1 and TRAF2, resulting in inhibition of caspase 8 [32]. Expression of both cIAP1 and cIAP2 are NF-κB dependent [32,33]. NF-κB also regulates X chromosome-linked IAP (XIAP) [34], which inhibits caspases 3, 7 and 9 [35,36]. Thyomocyte apoptosis induced by various death insults was suppressed in T cell-specific XIAP transgenic mice [37]. It is important to note that *Xiap*$^{-/-}$ mice showed no difference in caspase-mediated apoptosis, although cIAP1 and cIAP2 were up-regulated [38]. This suggests the existence

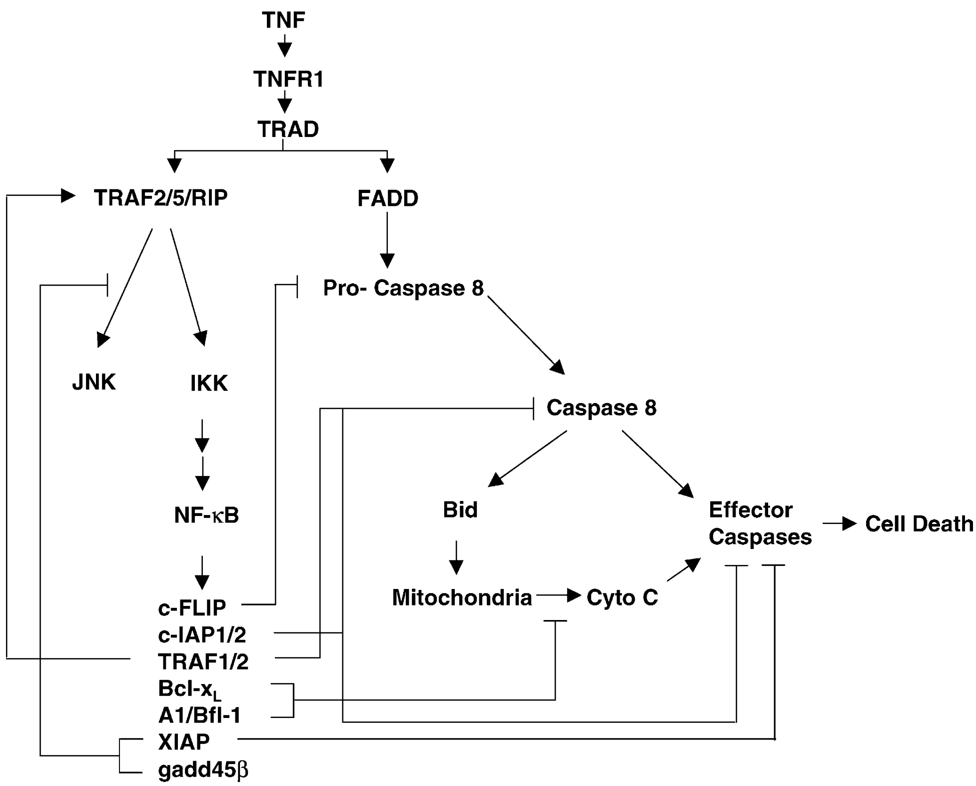

Fig. 2. Mechanisms behind the anti-apoptotic activity of NF-κB. NF-κB inhibits TNFα-induced apoptosis through induction of a variety of anti-apoptotic factors, including inhibitors of caspase activation (cFLIP) and action (cIAPs), anti-apoptotic Bcl-2 family members (A1, Bcl-X_L) and inhibitors of JNK activation (XIAP, Gadd45β).

of a compensatory mechanism. Overexpression of XIAP can inhibit TNF-α induced apoptosis [39]. As discussed later, XIAP may also mediate the anti-apoptotic function of NF-κB through inhibition of prolonged JNK activation [40] (F. Tang and A. Lin, unpublished data).

cFLIP is another NF-κB regulated inhibitor of apoptosis [41,42]. Since cFLIP interacts with FADD and pro-caspase 8 [43], it may inhibit apoptosis by interfering with activation of pro-caspase 8. In addition, cFLIP interacts with TRAF2 and receptor-interacting protein (RIP) [44], two adaptor proteins that are responsible for activation of IKK in the TNFR1 complex [45]. However, it is not clear whether the anti-apoptotic effect of cFLIP is mediated by augmenting NF-κB activation [41]. While *cFlip*$^{-/-}$ fibroblasts are hypersensitive to TNF-α or FasL induced apoptosis [41], ectopic expression of cFLIP prevents TNF-α-induced apoptosis in cells expressing a degradation-resistant form (super-repressor) of IκBα [42]. These findings suggest that cFLIP may be involved in mediating NF-κB anti-apoptotic effects.

NF-κB also regulates expression of several members of the Bcl-2 family. One of these apoptosis-regulating proteins is Al, which is a hematopoietic-specific Bcl-2 homologue [46]. In addition to suppression of apoptosis induced by IL-3 withdrawal and TNF-α [47], over-expression of A1 prevents mitochondrial depolarization, the release of cytochrome *c* and caspase 9 activation induced by DNA damaging agents such as *etoposide* [47]. Since *A1*$^{-/-}$ mice only exhibited increased apoptosis in neutrophils [47], its anti-apoptotic role may be limited to certain tissues. NF-κB may also regulate Bcl-X_L, an anti-apoptotic member of the Bcl-2 family. Expression of Bcl-X_L is enhanced by NF-κB inducers such as CD28 or CD40 ligation, PMA plus ionomycin, TNF-α and the HTLV1 Tax protein [48–51]. Conversely, Bcl-X_L overexpression rescues cells expressing the IκBα super-repressor from TNF-α induced apoptosis [51]. Bcl-2 itself may also be involved in the anti-apoptotic function of NF-κB. *c-Rel*$^{-/-}$*RelA*$^{-/-}$ B cells were defective in Bcl-2 expression and a Bcl-2 transgene partially prevented the apoptosis of such B cells engrafted to irradiated hosts [52]. However, it remains to be determined whether expression of Bcl-X_L or Bcl-2 is directly regulated by NF-κB. The anti-apoptotic function of NF-κB may also be mediated by its inhibition of the pro-apoptotic member Bax, albeit probably indirectly [53]. This inhibition might be important in the survival of cancer cells that exhibit constitutive NF-κB activity [53].

NF-κB can suppress cell death through regulation of anti-apoptotic and pro-apoptotic signaling pathways. As discussed earlier, NF-κB induces expression of both TRAF1 and TRAF2 [32]. This may augment activation of NF-κB since TRAF2 plays a critical role in TNF-α induced activation of the IKK and JNK pathways, which lead to NF-κB and AP-1 activation, respectively [45]. Indeed, *Traf2*$^{-/-}$ mice exhibit increased sensitivity to TNF-α induced apoptosis [54]. However, *Traf2*$^{-/-}$ fibroblasts are only partially deficient in NF-κB activation, but completely defective in TNF-α stimulated JNK activity [54]. The exact mechanism behind the increased sensitivity to TNF-α has yet to be determined.

The role of JNK activation in TNF-α induced apoptosis has been a highly controversial issue. Based on the results of transient transfection experiments using JNK activators or dominant negative forms of c-Jun, it was postulated that JNK activation is pro-apoptotic in TNF-α signaling [55]. However, dissection of TNF-α-stimulated signaling pathways did not support a pro-apoptotic role for JNK [25] or even suggested a weak anti-apoptotic function [56], while revealing the anti-apoptotic activity of NF-κB [25]. JNK activation may not be obligatory for TNF-α-induced apoptosis that depends on both extrinsic and intrinsic death pathways, although it is required for UV-induced apoptosis, which depends on the intrinsic death pathway [57]. Nevertheless, JNK may be a positive modulator of TNF-α induced apoptosis through an unknown mechanism [40]. A constitutively-activated JNK construct enhanced TNF-α induced apoptosis of *RelA*$^{-/-}$ cells, while having no apoptotic activity on its own [40]. Furthermore, JNK activation by TNF-α is highly transient in normal cells, but is prolonged in cells deficient in IKKβ or RelA, or expressing the IκBα super-repressor [40,58,59]. It is important to note that the loss of NF-κB activity did not alter the kinetics of JNK activation by interleukin 1 (IL-1) [40]. This suggests that NF-κB-mediated inhibition of JNK activation is specific for TNF-α, even though TNF-α and IL-1 share a great deal of resemblance and overlap in their activation of the JNK and NF-κB pathways.

The observations above suggest that NF-κB induces expression of a JNK inhibitor(s), which contributes to the anti-apoptotic function of NF-κB. Yet the identity of NF-κB-induced JNK inhibitor is not clear. Ectopic expression of XIAP, which is a caspase inhibitor, converted JNK activation from prolonged to transient in TNF-α treated *RelA*$^{-/-}$ cells and MCF-7 cells without affecting activation of p38 or extracellular signal-regulated kinase (ERK) [40] (F. Tang and A. Lin, unpublished data). Since caspase activation by death receptors can result in prolonged JNK activation [60], XIAP-mediated inhibition of JNK activation could be due to its inhibition of caspases. This is unlikely since TNF-α still induces prolonged JNK activation in *RelA*$^{-/-}$ cells even in the presence of the caspase inhibitor zVAD (G. Tang and A. Lin, unpublished data). Other candidates of NF-κB-induced JNK inhibitors include GADD45β [58], which was known as a cell cycle inhibitor [61]. However, both *Xiap*$^{-/-}$ and *Gadd45b*$^{-/-}$ mice are viable and have no massive apoptosis in liver associated with loss of NF-κB activity [38] (A. Fornace and D. Liebermann, personal communication). It is likely that NF-κB may inhibit JNK through induction of a group of inhibitors. In addition to identification of NF-κB-induced JNK inhibitors, future studies need to elucidate how and where the JNK signaling pathway is inhibited.

It remains to be determined whether inhibition of JNK activation by NF-κB affects apoptosis induced by other death signals. It is important to note that the role of JNK in apoptosis is greatly affected by many factors such as cell type, activation duration and the nature of death insult [62]. Furthermore, NF-κB can be pro-apoptotic under certain circumstances (see the description later). So the effect of NF-κB-mediated inhibition of JNK on apoptosis is likely influenced by other cellular regulators. Interestingly, another MAPK, p38, synergizes with NF-κB in the induction of anti-apoptotic genes in macrophages [63]. Inhibition of either IKKβ or p38 in macrophages renders them very sensitive to lipopolysaccharide and lipoteichoic acid, bacterial cell wall ingredients that normally are very potent macrophage activators [63].

The take home message is that NF-κB suppresses apoptosis through induction of multiple target genes coding for inhibitors of the death pathways, extrinsic or intrinsic, or other signaling pathways that regulate apoptosis. The ultimate effect of NF-κB activation on apoptosis depends on the cell type and the particular situation in which they are examined. It is curious to note that none of the mice deficient in a singular NF-κB-inducible anti-apoptotic factor have the massive liver apoptosis as exhibited by *RelA*$^{-/-}$ or *Ikk*β$^{-/-}$ mice. This suggests that multiple NF-κB induced inhibitors are involved in suppression of TNF-α induced apoptosis or the critical factor has yet to be identified.

5. Death by eliminating NF-κB activation

Considering that NF-κB-dependent gene expression is such a major obstacle on the road to apoptosis, its activity has to be circumvented for apoptotic signals to induce cell death. Indeed, many components of the NF-κB pathway are targeted by either caspase-mediated proteolysis or inhibited by other mechanisms, resulting in termination of the anti-apoptotic activity of NF-κB.

Both RIP and TRAF-2, which are involved in TNF-α-induced NF-κB activation, are substrates of caspases. RIP is cleaved into two major fragments by caspase 8, eliminating signaling from RIP to IKK [64]. The N-terminal truncated fragment of RIP also enhances the association between TNFR1, TRADD and FADD but inhibits NF-κB activation upon over-expression, thereby promoting TNF-α induced apoptosis [64]. Proteolysis of TRAF2 or its sequestration into a non-soluble cellular compartment in response to ligation of CD30 (a member of the TNFR family) may impair the anti-apoptotic activity of NF-κB, resulting in increased sensitivity to TNF-α induced apoptosis [65]. TRAF1 is also cleaved by caspase 8 during TNF-α- or Fas-induced apoptosis [66,67]. The C-terminal fragment of TRAF1 inhibited NF-κB activation by co-transfected TRAF2 or TNFR1 and promoted apoptosis induced by TNF-α and Fas ligation [66,67], probably by disrupting the interaction of TRAF1 with TRAF2, as well as with cIAP1 and cIAP2 [67].

IKKβ is essential for NF-κB activation by TNF-α, while IKKα is dispensable [7]. During TNF-α or FasL induced apoptosis, IKKβ, but not IKKα or IKKγ, is cleaved by caspase 3-related caspases [68]. Cleavage of IKKβ results in elimination of its enzymatic activity [68]. One of the proteolytic fragments, IKKβ (1–546), is able to inhibit IKK activity and promotes TNF-α induced apoptosis when over-expressed [68]. Most importantly, over-expression of a caspase-resistant IKKβ mutant suppresses apoptosis induced by TNF-α, as well as Fas ligation [68]. Therefore, the proteolysis of IKKβ may be a critical event that determines life or death of cells exposed to TNF-α or Fas ligation. How does the caspase-resistant IKKβ mutant inhibit apoptosis? It was found that in wild type cells, a singular TNF-α treatment resulted in repeated activation of IKK, which was eventually diminished following the proteolysis of IKKβ. In contrast, this repeated activation of IKK lasted for an extended period of time in cells expressing the caspase-resistant IKKβ mutant [68]. This may allow accumulation of NF-κB-induced anti-apoptotic proteins, thereby inhibiting apoptosis [68]. Another possibility is that prolonged IKK activation in the presence of the caspase-resistant IKKβ mutant may prevent proteolysis of other components in the NF-κB pathway by caspases, as discussed later.

IκBα is also targeted by caspase-mediated proteolysis. While ubiquitination-mediated proteolysis of IκBα by the 26S proteosome leads to NF-κB activation [7], proteolysis of IκBα by caspase 3 inhibited NF-κB activation instead. This is because the proteolysis generated a N-terminal truncated molecule that still binds NF-κB and is resistant to TNF-α-induced phosphorylation and degradation [69]. Interestingly, phosphorylation of IκBα by IKK or replacement of Ser32 and Ser36 with glutamates to mimic their phosphorylation prevented caspase-mediated proteolysis [70]. This suggests that phosphorylation of IκBα by IKK not only triggers proteosome-dependent proteolysis of IκBα to release NF-κB, but it also prevents the generation of a super-repressor-like fragment of IκBα by caspases.

NF-κB itself is also a target of caspases. The proteolysis of RelA by caspases occurs during cell death induced by growth factor withdrawal, resulting a C-terminal truncated RelA that still binds DNA but cannot activate transcription [71]. This converts RelA into its own inhibitor. The importance of RelA proteolysis in termination of NF-κB activation is evident by the observation that expression of a caspase-resistant RelA mutant protects cells against apoptosis caused by growth factor withdrawal [71].

Another way to terminate NF-κB activation during apoptosis is to destroy its anti-apoptotic gene products by proteolysis. Indeed, cells utilize caspases to cleave NF-κB-induced anti-apoptotic inhibitors, including cIAP1, XIAP and Bcl-X_L [36,72–74]. A common theme is that in addition to abrogation of their anti-apoptotic ability, proteolysis of the anti-apoptotic inhibitors by caspases usually generates pro-apoptotic fragments [36,72–74]. This probably provides a more efficient way to eliminate the protection conveyed by NF-κB.

The anti-apoptotic activity of NF-κB can also be turned off by the ingestion of apoptotic bodies by macrophages [75]. The uptake of apoptotic bodies switches off the production of TNF-α and other inflammatory cytokines and induces the expression of anti-inflammatory cytokines, such as TGFβ. The mechanism underlying this phenomenon has yet to be determined. In addition, NF-κB activation can be inhibited by YopJ, an ubiquitin-like protein protease produced by *Yersinia pestis*, the causative agent of bubonic plague [76]. By binding to MAP kinase (MAPK) kinase kinases (MAP3Ks) as well as to IKK, YopJ inhibits their activation [76]. This may account for the ability of *Y. pestis* to suppress the host inflammatory response and to induce apoptosis in host cells [77]. Recently, it was found that the lethal toxin of *Bacillus anthracis* can trigger the apoptosis of activated macrophages by preventing the activation of p38, whose activity is needed for the induction of NF-κB-dependent anti-apoptotic genes [63]. This strategy allows *B. anthracis* to escape detection by the innate immune system, just like *Y. pestis*.

6. A pro-apoptotic transcriptional factor?

Although overwhelming evidence, especially those obtained form analysis of knockout mice, strongly support the anti-apoptotic function of NF-κB, a few reports suggest that NF-κB may also contribute to induction of pro-apoptotic molecules, including death receptor 6 (DR6), a member of the TNFR family [78], DR4 and DR5 [79], as well as Fas [80]. However, in each case NF-κB concurrently induced expression of anti-apoptotic molecules that neutralize the pro-apoptotic activity. Thus, it has yet to be determined whether induction of pro-apoptotic molecules by NF-κB is physiologically relevant.

7. Targeting NF-κB for cancer therapy

The anti-apoptotic activity of NF-κB is likely to play a critical role in oncogenesis (Fig. 3). Constitutive NF-κB activity has been found in breast, prostate, colorectal and ovarian cancers, and certain forms of leukemia and lymphoma [9,10]. The ability of NF-κB to inhibit apoptosis,

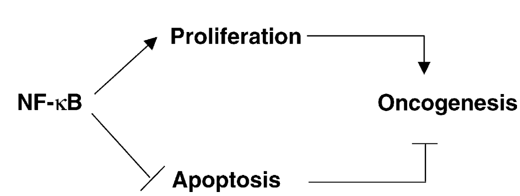

Fig. 3. NF-κB is a key regulator in oncogenesis. Two modes of NF-κB's role in oncogenesis. In addition to promotion of cell growth, activation of NF-κB inhibits apoptosis, thereby contributing to oncogenesis.

as well as promoting cell proliferation makes it an attractive target for cancer therapy. Indeed, inhibition of NF-κB triggered apoptosis in certain leukaemias and lymphomas, suggesting that its activity is essential for growth and/or survival of these cancer cells [81–83]. In addition, inhibition of NF-κB resulted in increased sensitivity of tumor cells to cancer therapy-induced apoptosis, especially in response to chemotherapeutic drugs and γ-radiation [30,84]. However, inhibition of NF-κB does not always promote apoptosis in tumor cells. It is possible that malignant tumors, which possess numerous genetic alterations, may no longer solely depend on NF-κB activity for survival [9]. Another possibility is that the role of NF-κB in apoptosis may depend on tumor type and the nature of death insults. Accumulating evidence suggests that the anti-apoptotic activity of NF-κB may be required at the early stage of transformation. For instance, the occurrence of colorectal and gastric tumors is suppressed by several anti-inflammatory drugs and dietary components. The list includes aspirin and other non-steroidal anti-inflammatory drugs (NSAIDs), curcumin, flavonoids and resveratrol [85–88], which also inhibit NF-κB activation. It is plausible that these agents may prevent cancer by inhibiting NF-κB activity. Further studies are necessary to substantiate this hypothesis. Since the majority of cells with DNA alterations or damage are eliminated by apoptosis during the transformation crisis, NF-κB activity might be required for inhibiting this transformation-related apoptosis [89,90]. Mechanisms apart, the anti-apoptotic activity of NF-κB is likely to contribute to oncogenesis.

A major concern about utilizing NF-κB inhibition as a therapeutic strategy in cancer prevention and treatment is the issue of selectivity. NF-κB is required for many fundamental cellular activities including immune responses. Any successful NF-κB inhibitor used for the purpose of cancer therapy should avoid suppressing the immune system, which could be important for cancer surveillance and treatment. Recent progress in understanding the mechanisms that regulate NF-κB activity may provide strategies in designing selective inhibitors of NF-κB activation. One such strategy is to design IKKα specific inhibitors. Although IKKβ is essential for mediating NF-κB activation in innate immune responses, IKKα is dispensable [7]. Thus, IKKα specific inhibitors may selectively block NF-κB activation in certain cancers without affecting the innate immune system. The anti-apoptotic activity of NF-κB is mediated through induction of genes whose products inhibit either extrinsic or intrinsic death pathways or other pro- or anti-apoptosis signaling pathways [12]. Therefore certain NF-κB target genes that are crucial for survival of particular cancer cells may be promising targets. A better understanding of the NF-κB biology should provide a molecular basis to eliminate NF-κB activity for the purpose of cancer prevention and treatment.

In conclusion, NF-κB is a key player in oncogenesis. It not only promotes proliferation but also inhibits apoptosis through induction of death suppressing genes. From the cancer therapy point of view, the key issue is to disassociate the anti-apoptotic activity of NF-κB from its functions in other cellular events such as immune responses. A major challenge for future research is to completely elucidate the mechanisms that are involved in regulation of NF-κB activation and the physiological relevance of NF-κB target genes in different cells. This may lead to the birth of therapeutic inhibitors that selectively block NF-κB activation in cancer cells without affecting NF-κB's functions in normal cells.

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References

[1] Bishop JM. Molecular themes in oncogenesis. Cell 1991;64:235–48.
[2] Hunter T. Cooperation between oncogenes. Cell 1991;64:249–70.
[3] Cantley LC, et al. Oncogenes and signal transduction. Cell 1991;64:281–302.
[4] Fisher DE. Apoptosis in cancer therapy: crossing the threshold. Cell 1994;78:539–42.
[5] Evan GI, Vousden KH. Proliferation, cell cycle and apoptosis in cancer. Nature 2001;411:342–8.
[6] Johnstone RW, Ruefli AA, Lowe SW. Apoptosis: a link between cancer genetics and chemotherapy. Cell 2002;108:153–64.
[7] Karin M, Ben-Neriah Y. Phosphorylation meets ubiquitination: the control of NF-κB activity. Annu Rev Immunol 2000;18:621–63.
[8] Ghosh S, Karin M. Missing pieces in the NF-κB puzzle. Cell 2002;109(Suppl):S81–96.
[9] Baldwin AS. Control of oncogenesis and cancer therapy resistance by the transcription factor NF-κB. J Clin Invest 2001;107:241–6.
[10] Karin M, Cao Y, Greten FR, Li ZW. NF-κB in cancer: from innocent bystander to major culprit. Nat Rev Cancer 2002;2:301–10.
[11] Gilmore TD. Multiple mutations contribute to the oncogenicity of the retroviral oncoprotein v-Rel. Oncogene 1999;18:6925–37.
[12] Karin M, Lin A. NF-κB at the crossroads of life and death. Nat Immunol 2002;3:221–7.
[13] Baldwin Jr AS. The NF-κB and IκB proteins: new discoveries and insights. Annu Rev Immunol 1996;14:649–83.
[14] Li ZW, et al. The IKKβ subunit of IκB kinase (IKK) is essential for nuclear factor kappa B activation and prevention of apoptosis. J Exp Med 1999;189:1839–45.
[15] Li Q, Van Antwerp D, Mercurio F, Lee KF, Verma IM. Severe liver degeneration in mice lacking the IκB kinase 2 gene. Science 1999;284:321–5.
[16] Tanaka M, et al. Embryonic lethality, liver degeneration, and impaired NF-κB activation in IKKβ-deficient mice. Immunity 1999;10:421–9.
[17] Senftleben U, Li ZW, Baud V, Karin M. IKKβ is essential for protecting T cells from TNF-α-induced apoptosis. Immunity 2001;14:217–30.
[18] Solan NJ, Miyoshi H, Carmona EM, Bren GD, Paya C. RelB cellular regulation and transcriptional activity are regulated by p100. J Biol Chem 2002;277:1405–18.
[19] Zhong H, SuYang H, Erdjument-Bromage H, Tempst P, Ghosh S. The transcriptional activity of NF-κB is regulated by the IκB-associated

PKAc subunit through a cyclic AMP-independent mechanism. Cell 1997;89:413–24.
[20] Sizemore N, Leung S, Stark GR. Activation of phosphatidylinositol 3-kinase in response to interleukin-1 leads to phosphorylation and activation of the NF-κB p65/RelA subunit. Mol Cell Biol 1999;19:4798–805.
[21] Madrid LV, et al. Akt suppresses apoptosis by stimulating the transactivation potential of the RelA/p65 subunit of NF-κB. Mol Cell Biol 2000;20:1626–38.
[22] Beg AA, Sha WC, Bronson RT, Ghosh S, Baltimore D. Embryonic lethality and liver degeneration in mice lacking the RelA component of NF-κB. Nature 1995;376:167–70.
[23] Beg AA, Baltimore D. An essential role for NF-κB in preventing TNF-α-induced cell death. Science 1996;274:782–4.
[24] Van Antwerp DJ, Marti SJ, Kafri T, Green DR, Verma IM. Suppression of TNF-α-induced apoptosis by NF-κB. Science 1996;274:787–9.
[25] Liu ZG, Hsu H, Goeddel DV, Karin M. Dissection of TNF receptor 1 effector functions: JNK activation is not linked to apoptosis while NF-κB activation prevents cell death. Cell 1996;87:565–76.
[26] Makris C, et al. Female mice heterozygous for IKKγ/NEMO deficiencies develop a dermatopathy similar to the human X-linked disorder incontinentia pigmenti. Mol Cell 2000;5:969–79.
[27] Alcamo E, et al. Targeted mutation of TNF receptor I rescues the RelA-deficient mouse and reveals a critical role for NF-κB in leukocyte recruitment. J Immunol 2001;167:1592–600.
[28] Doi TS, et al. Absence of tumor necrosis factor rescues RelA-deficient mice from embryonic lethality. Proc Natl Acad Sci USA 1999;96:2994–9.
[29] Horwitz BH, Scott ML, Cherry SR, Bronson RT, Baltimore D. Failure of lymphopoiesis after adoptive transfer of NF-κB-deficient fetal liver cells. Immunity 1997;6:765–72.
[30] Wang CY, Cusack Jr JC, Liu R, Baldwin Jr AS. Control of inducible chemoresistance: enhanced anti-tumor therapy through increased apoptosis by inhibition of NF-κB. Nat Med 1999;5:412–7.
[31] Deveraux QL, et al. IAPs block apoptotic events induced by caspase-8 and cytochrome *c* by direct inhibition of distinct caspases. EMBO J 1998;17:2215–23.
[32] Wang CY, Mayo MW, Korneluk RG, Goeddel DV, Baldwin Jr AS. NF-κB antiapoptosis: induction of TRAF1 and TRAF2 and c-IAP1 and c-IAP2 to suppress caspase-8 activation. Science 1998;281:1680–3.
[33] Chu ZL, et al. Suppression of tumor necrosis factor-induced cell death by inhibitor of apoptosis c-IAP2 is under NF-κB control. Proc Natl Acad Sci USA 1997;94:10057–62.
[34] Liston P, et al. Suppression of apoptosis in mammalian cells by NAIP and a related family of IAP genes. Nature 1996;379:349–53.
[35] Takahashi R, et al. A single BIR domain of XIAP sufficient for inhibiting caspases. J Biol Chem 1998;273:7787–90.
[36] Deveraux QL, et al. Cleavage of human inhibitor of apoptosis protein XIAP results in fragments with distinct specificities for caspases. EMBO J 1999;18:5242–51.
[37] Conte D, Liston P, Wong JW, Wright KE, Korneluk RG. Thymocyte-targeted overexpression of XIAP transgene disrupts T lymphoid apoptosis and maturation. Proc Natl Acad Sci USA 2001;98:5049–54.
[38] Harlin H, Reffey SB, Duckett CS, Lindsten T, Thompson CB. Characterization of XIAP-deficient mice. Mol Cell Biol 2001;21:3604–8.
[39] Stehlik C, et al. Nuclear factor (NF)-kappaB-regulated X-chromosome-linked IAP gene expression protects endothelial cells from tumor necrosis factor alpha-induced apoptosis. J Exp Med 1998;188:211–6.
[40] Tang G, et al. Inhibition of JNK activation through NF-κB target genes. Nature 2001;414:313–7.
[41] Yeh WC, et al. Requirement for Casper (c-FLIP) in regulation of death receptor-induced apoptosis and embryonic development. Immunity 2000;12:633–42.
[42] Kreuz S, Siegmund D, Scheurich P, Wajant H. NF-κB inducers upregulate cFLIP, a cycloheximide-sensitive inhibitor of death receptor signaling. Mol Cell Biol 2001;21:3964–73.
[43] Irmler M, et al. Inhibition of death receptor signals by cellular FLIP. Nature 1997;388:190–5.
[44] Shu HB, Halpin DR, Goeddel DV. Casper is a FADD- and caspase-related inducer of apoptosis. Immunity 1997;6:751–63.
[45] Baud V, Karin M. Signal transduction by tumor necrosis factor and its relatives. Trends Cell Biol 2001;11:372–7.
[46] Lin EY, Orlofsky A, Berger MS, Prystowsky MB. Characterization of A1, a novel hemopoietic-specific early-response gene with sequence similarity to bcl-2. J Immunol 1993;151:979–88.
[47] Wang CY, Guttridge DC, Mayo MW, Baldwin Jr AS. NF-κB induces expression of the Bcl-2 homologue A1/Bfl-1 to preferentially suppress chemotherapy-induced apoptosis. Mol Cell Biol 1999;19:5923–9.
[48] Lee HH, Dadgostar H, Cheng Q, Shu J, Cheng G. NF-κB-mediated up-regulation of Bcl-x and Bfl-1/A1 is required for CD40 survival signaling in B lymphocytes. Proc Natl Acad Sci USA 1999;96:9136–41.
[49] Tsukahara T, et al. Induction of Bcl-x(L) expression by human T-cell leukemia virus type 1 Tax through NF-κB in apoptosis-resistant T-cell transfectants with Tax. J Virol 1999;73:7981–7.
[50] Khoshnan A, et al. The NF-κB cascade is important in Bcl-xL expression and for the anti-apoptotic effects of the CD28 receptor in primary human CD4+ lymphocytes. J Immunol 2000;165:1743–54.
[51] Chen C, Edelstein LC, Gelinas C. The Rel/NF-κB family directly activates expression of the apoptosis inhibitor Bcl-x(L). Mol Cell Biol 2000;20:2687–95.
[52] Grossmann M, et al. The anti-apoptotic activities of Rel and RelA required during B-cell maturation involve the regulation of Bcl-2 expression. EMBO J 2000;19:6351–60.
[53] Bentires-Alj M, et al. Inhibition of the NF-κB transcription factor increases Bax expression in cancer cell lines. Oncogene 2001;20:2805–13.
[54] Lee SY, et al. TRAF2 is essential for JNK but not NF-κB activation and regulates lymphocyte proliferation and survival. Immunity 1997;7:703–13.
[55] Verheij M, et al. Requirement for ceramide-initiated SAPK/JNK signalling in stress-induced apoptosis. Nature 1996;380:75–9.
[56] Natoli G, et al. Activation of SAPK/JNK by TNF receptor 1 through a noncytotoxic TRAF2-dependent pathway. Science 1997;275:200–3.
[57] Tournier C, et al. Requirement of JNK for stress-induced activation of the cytochrome *c*-mediated death pathway. Science 2000;288:870–4.
[58] De Smaele E, et al. Induction of gadd45β by NF-κB downregulates pro-apoptotic JNK signalling. Nature 2001;414:308–13.
[59] Javelaud D, Besancon F. NF-κB activation results in rapid inactivation of JNK in TNF-α-treated Ewing sarcoma cells: a mechanism for the anti-apoptotic effect of NF-κB. Oncogene 2001;20:4365–72.
[60] Lenczowski JM, et al. Lack of a role for Jun kinase and AP-1 in Fas-induced apoptosis. Mol Cell Biol 1997;17:170–81.
[61] Zhan Q, et al. Association with Cdc2 and inhibition of Cdc2/Cyclin B1 kinase activity by the p53-regulated protein Gadd45. Oncogene 18:2892–900.
[62] Lin A. Activation of the JNK signaling pathway: breaking the brake on apoptosis. BioEssays, in press.
[63] Park JM, Greten FR, Li Z-W, Karin M. Macrophage apoptosis by anthrax lethal factor and p38 kinase inhibition. Science 2002, in press.
[64] Lin Y, Devin A, Rodriguez Y, Liu ZG. Cleavage of the death domain kinase RIP by caspase-8 prompts TNF-induced apoptosis. Genes Dev 1999;13:2514–26.
[65] Arch RH, Gedrich RW, Thompson CB. Translocation of TRAF proteins regulates apoptotic threshold of cells. Biochem Biophys Res Commun 2000;272:936–45.

[66] Leo E, et al. TRAF1 is a substrate of caspases activated during tumor necrosis factor receptor-alpha-induced apoptosis. J Biol Chem 2001;276:8087–93.
[67] Schwenzer R, et al. The human tumor necrosis factor (TNF) receptor-associated factor 1 gene (TRAF1) is up-regulated by cytokines of the TNF ligand family and modulates TNF-induced activation of NF-κB and c-Jun N-terminal kinase. J Biol Chem 1999;274:19368–74.
[68] Tang G, Yang J, Minemoto Y, Lin A. Blocking caspase-3-mediated proteolysis of IKKβ suppresses TNF-α-induced apoptosis. Mol Cell 2001;8:1005–16.
[69] Reuther JY, Baldwin Jr AS. Apoptosis promotes a caspase-induced amino-terminal truncation of IκBα that functions as a stable inhibitor of NF-κB. J Biol Chem 1999;274:20664–70.
[70] Barkett M, Xue D, Horvitz HR, Gilmore TD. Phosphorylation of IκBα inhibits its cleavage by caspase CPP32 in vitro. J Biol Chem 1997;272:29419–22.
[71] Levkau B, et al. xIAP induces cell-cycle arrest and activates nuclear factor-kappa B: new survival pathways disabled by caspase-mediated cleavage during apoptosis of human endothelial cells. Circ Res 2001;88:282–90.
[72] Clem RJ, et al. c-IAP1 is cleaved by caspases to produce a proapoptotic C-terminal fragment. J Biol Chem 2001;276:7602–8.
[73] Clem RJ, et al. Modulation of cell death by Bcl-XL through caspase interaction. Proc Natl Acad Sci USA 1998;95:554–9.
[74] Fujita N, Nagahashi A, Nagashima K, Rokudai S, Tsuruo T. Acceleration of apoptotic cell death after the cleavage of Bcl-XL protein by caspase-3-like proteases. Oncogene 1998;17:1295–304.
[75] Fadok VA, et al. Macrophages that have ingested apoptotic cells in vitro inhibit proinflammatory cytokine production through autocrine/paracrine mechanisms involving TGF-β, PGE2, and PAF. J Clin Invest 1998;101:890–8.
[76] Orth K, et al. Disruption of signaling by Yersinia effector YopJ, a ubiquitin-like protein protease. Science 2000;290:1594–7.
[77] Mills SD, et al. Yersinia enterocolitica induces apoptosis in macrophages by a process requiring functional type III secretion and translocation mechanisms and involving YopP, presumably acting as an effector protein. Proc Natl Acad Sci USA 1997;94:12638–43.
[78] Kasof GM, et al. Tumor necrosis factor-alpha induces the expression of DR6, a member of the TNF receptor family, through activation of NF-κB. Oncogene 2001;20:7965–75.
[79] Ravi R, et al. Regulation of death receptor expression and TRAIL/Apo2L-induced apoptosis by NF-κB. Nat Cell Biol 2001;3:409–16.
[80] Zheng Y, et al. NF-κB RelA (p65) is essential for TNF-α-induced fas expression but dispensable for both TCR-induced expression and activation-induced cell death. J Immunol 2001;166:4949–57.
[81] Furman RR, Asgary Z, Mascarenhas JO, Liou HC, Schattner EJ. Modulation of NF-κB activity and apoptosis in chronic lymphocytic leukemia B cells. J Immunol 2000;164:2200–6.
[82] Bargou R, et al. Constitutive nuclear factor-kappaB-RelA activation is required for proliferation and survival of Hodgkin's disease tumor cells. J Clin Invest 1997;100:2961–9.
[83] Cahir-McFarland E, Kieff E. NF-κB inhibition in EBV-transformed lymphoblastoid cell lines. Rec Results Cancer Res 2002;159:44–8.
[84] Wang CY, Mayo MW, Baldwin Jr AS. TNF- and cancer therapy-induced apoptosis: potentiation by inhibition of NF-κB. Science 1996;274:784–7.
[85] Kopp E, Ghosh S. Inhibition of NF-κB by sodium salicylate and aspirin. Science 1994;265:956–9.
[86] Yin MJ, Yamamoto Y, Gaynor RB. The anti-inflammatory agents aspirin and salicylate inhibit the activity of IκB kinase β. Nature 1998;396:77–80.
[87] Tsai SH, Liang YC, Lin-Shiau SY, Lin JK. Suppression of TNF-α-mediated NF-κB activity by myricetin and other flavonoids through downregulating the activity of IKK in ECV304 cells. J Cell Biochem 1999;74:606–15.
[88] Jobin C, et al. Curcumin blocks cytokine-mediated NF-kappa B activation and proinflammatory gene expression by inhibiting inhibitory factor I-kappa B kinase activity. J Immunol 1999;163:3474–83.
[89] Mayo MW, et al. Requirement of NF-κB activation to suppress p53-independent apoptosis induced by oncogenic Ras. Science 1997;278:1812–5.
[90] Sovak MA, et al. Aberrant nuclear factor-kappaB/Rel expression and the pathogenesis of breast cancer. J Clin Invest 1997;100:2952–60.